

Final Project presentation

- PROJECT: City Pollution Monitoring

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Meet the team



*Adrian
Iannantuono*



Drishya Suresh



Max Shui



Myron Soares



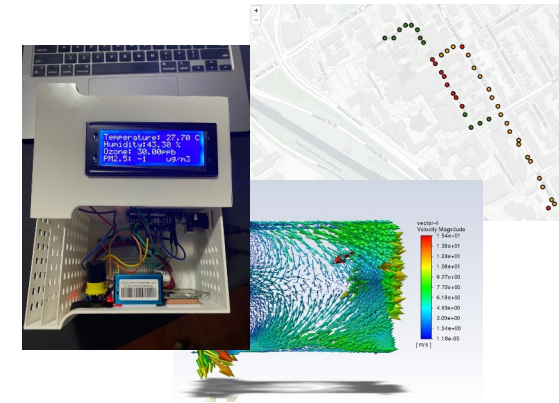
Tarin Sultana

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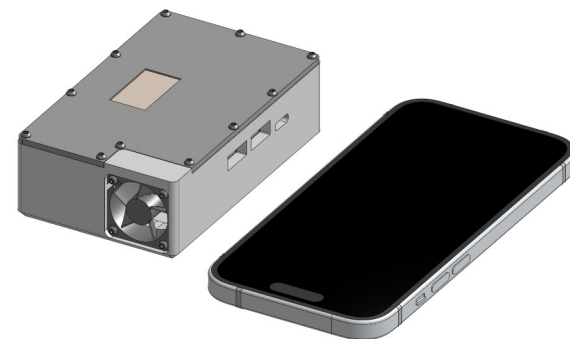
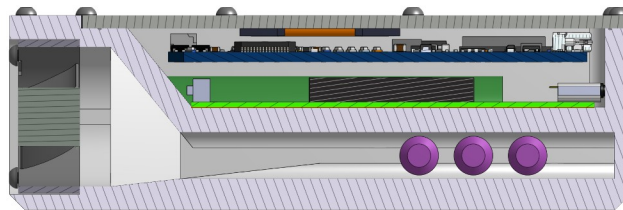
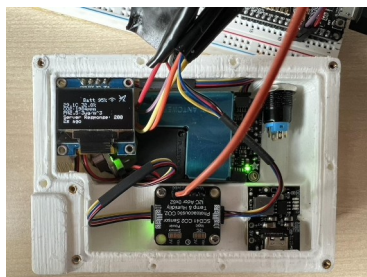
Initial Problem Statements

- The previous iteration of this project was a device mounted on an e-bike powered via an external battery and user friendly.
- The client requested for a reduction in size to improve its usability and also for easy data interpretation and visualization to the user.
- For the device to be functional through all parts of the year, durability in different weather conditions was to be kept in mind when designing the product.
- As socially responsible engineers to the environment, the development and research for this project was to be done while keeping the UNSDG goals in mind.



Initial Solutions

- Our initial solution was designed keeping in mind functionality toward the user.
- To help improve sensor accuracy we added a small fan to improve air flow over the sensors.
- The next step was to have an internal power bank to remove the need for external power and incentivize the product for the end user.
- Another step taken was to improve form factor, which was done by selection of miniature components and efficient packing design saving space and a great reduction in size.

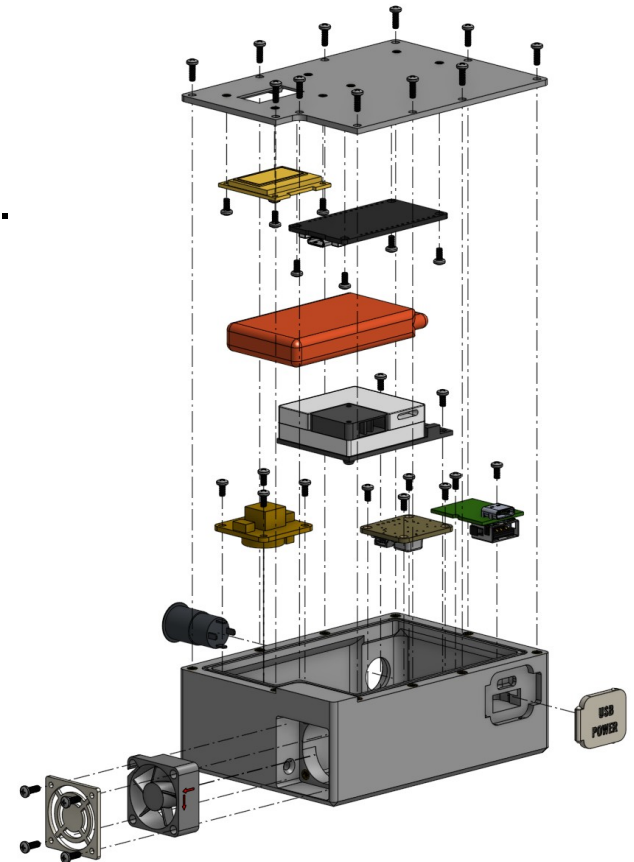
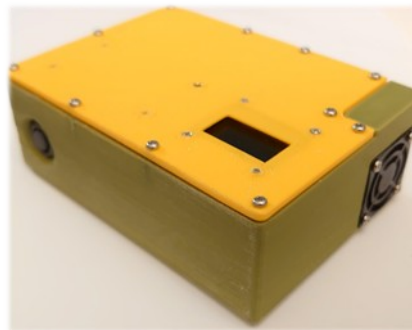
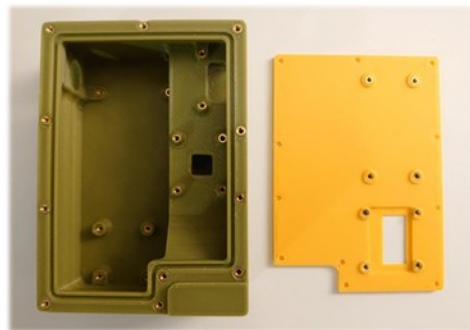


Problems Faced

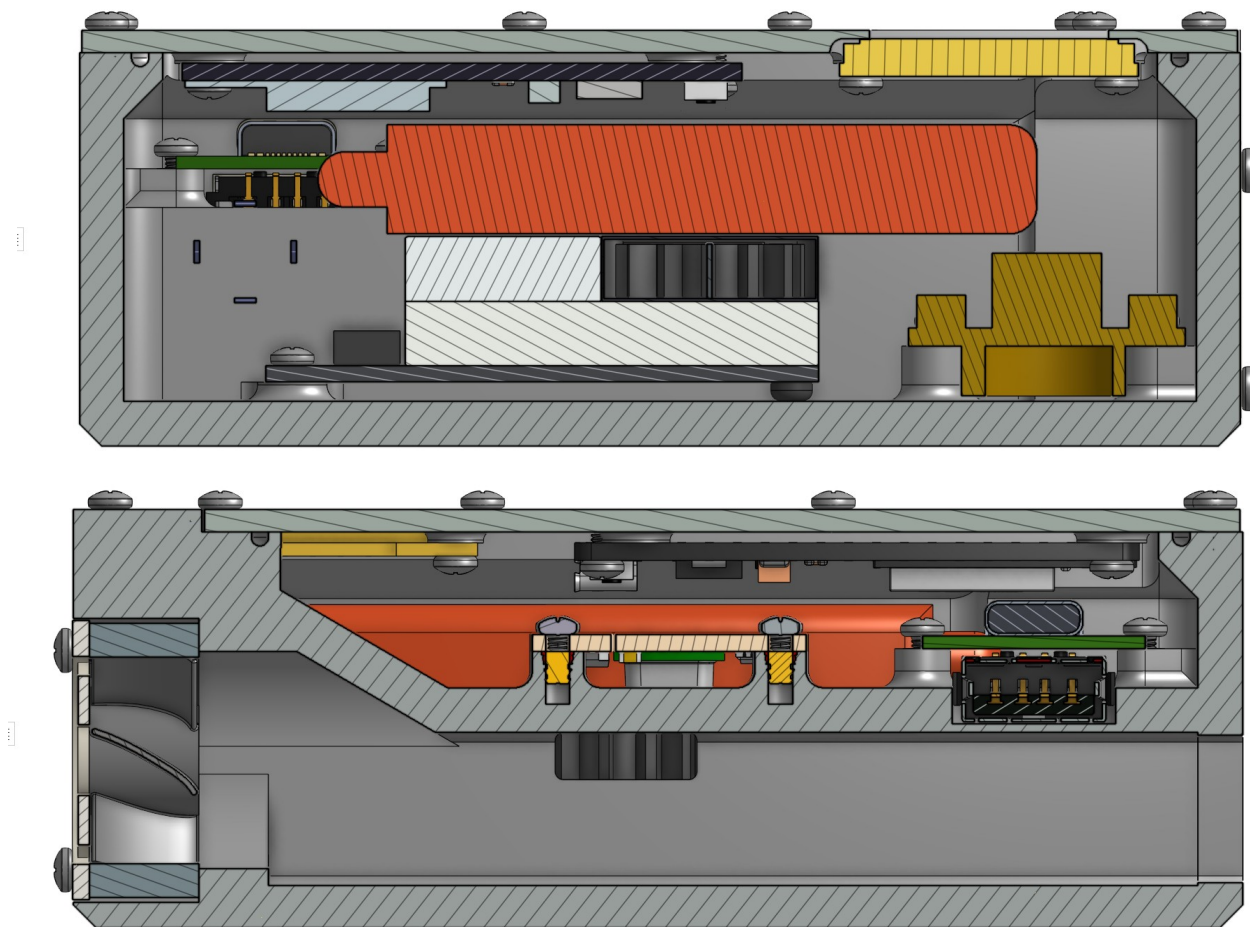
Problem faced	Response
How to incentivize volunteers to use the app?	Make the module usable as a portable power bank
How to get true and accurate readings for the sensors?	We incorporated a fan to push air through the channel for more accurate readings.
How to make it easier for the user to carry it around?	Implement of a standard VESA hole pattern in the housing that allows the attachment of a variety of mounts. The module can be attached to any vehicle or backpack using these attachments.

Final Solution concept and design (Mechanical)

- Final component packaging optimized for efficiency and ease of assembly in accordance with final component selection.
- Overall unit increased in size by about 8.6% from the original design.
- Relocation of GPS module for better reception.
- Ingress protection features improved
- Ergonomics improved.



Final Solution concept and design (Mechanical)



Final Solution concept and design (Electrical)

Microcontroller

- NodeMCU Amica with WiFi running the Arduino bootloader

Includes 3 sensors and 1 OLED Screen. All connected through the I²C protocol

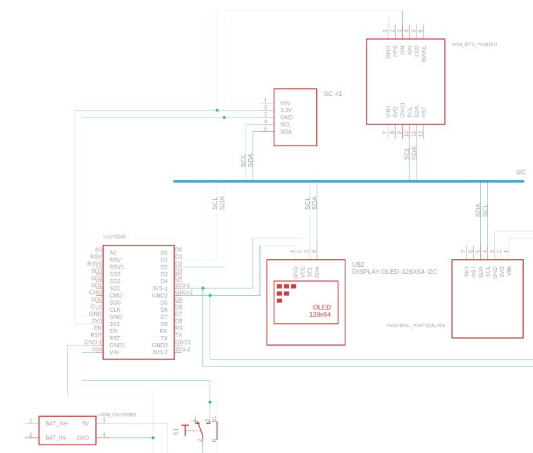
- Adafruit SCD-41 (Temperature, Humidity and CO₂)
- Adafruit PMSA003I (PM2.5)
- Adafruit PA1010D (GPS)

Li-ion battery and charger

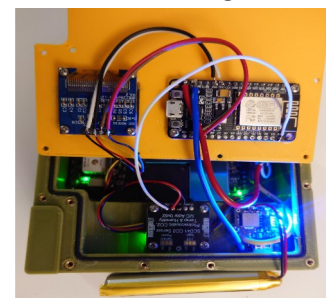
- Charges the battery via USB-C and can charge external devices through USB-A (max 2A)

Power Consumption:

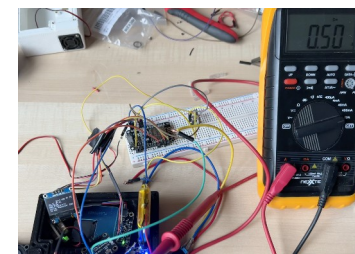
- 500mA peak (Assuming worst case measurement) = ~6h with 3000 mAh battery



Final Electrical Design Schematic



Final Electrical Hookup



Measuring Power Consumption

UI and Device Operation (Arduino)

- POST request containing collected data is sent every 10s to the app



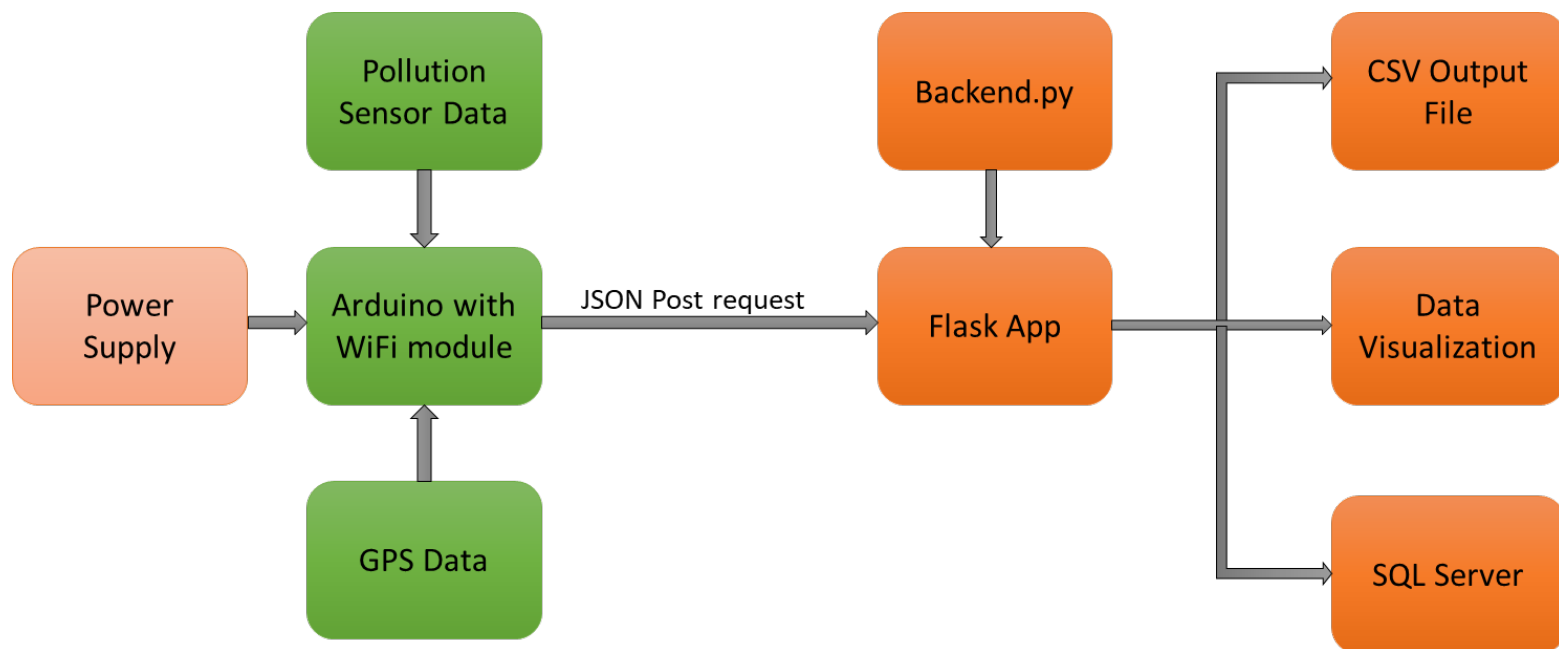
OLED display showing recorded data

TIME	REQUEST	RESPONSE
Today, 4:03 pm	POST /	Default
> REQUEST HEADERS		
> REQUEST BODY		
<pre>{ "date_measured": "2023-11-21", "time_measured": "16:03:29", "latitude": "45.256371N", "longitude": "75.413597W", "co2": "13331.00", "temperature": "4.11", "humidity": "39.17", "pm25": "3.00" }</pre>		
> RESPONSE HEADERS		
> RESPONSE BODY		

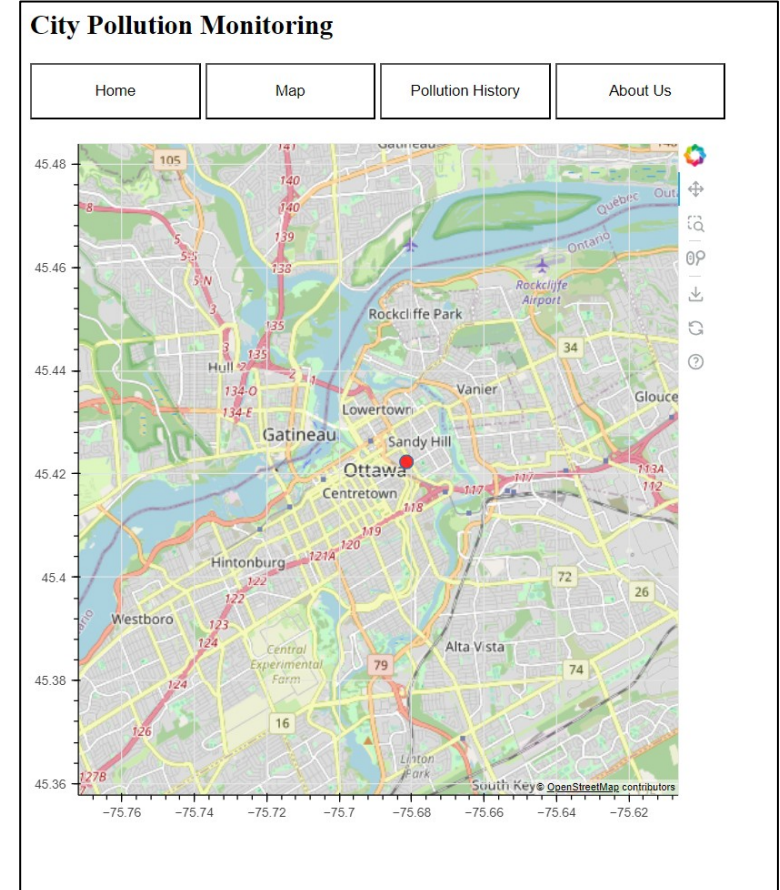
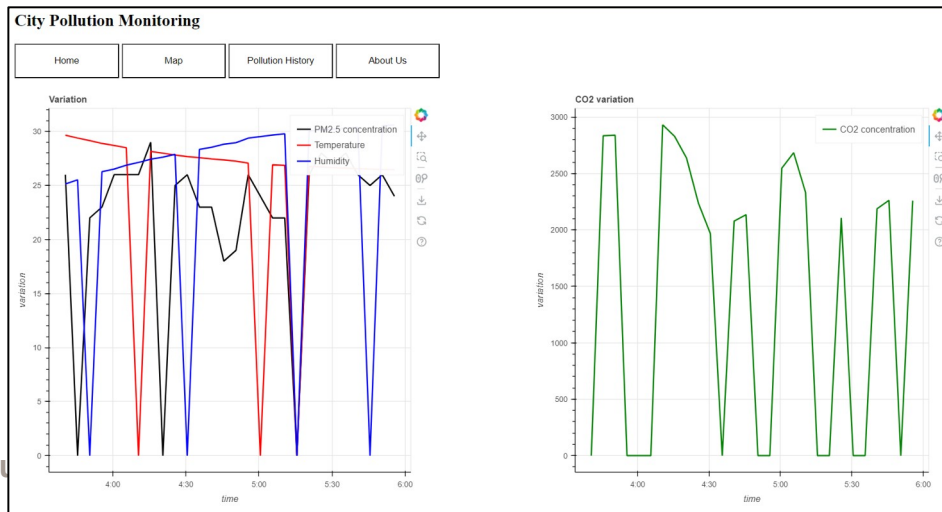
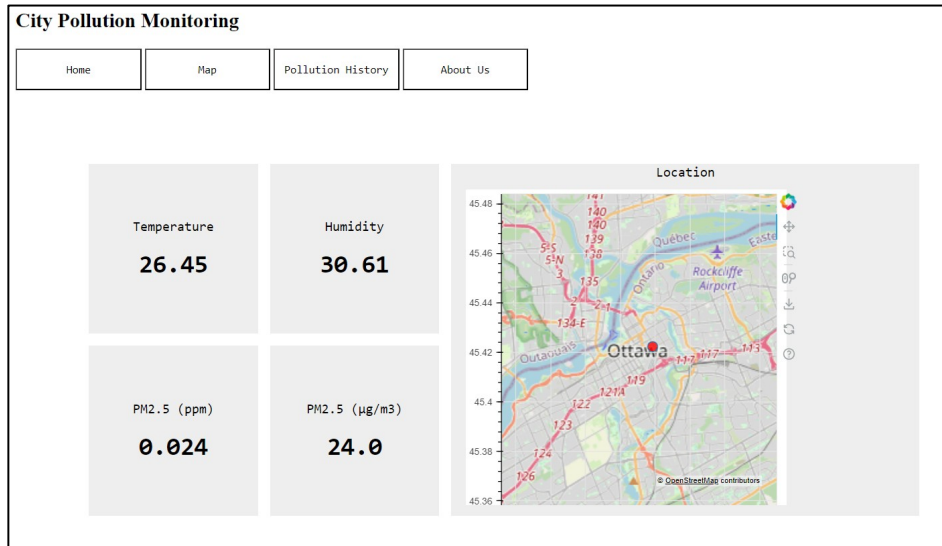
POST Request Received by Server From Device

Final Solution concept and design (Software)

- Web app made using Python (flask) and SQL Server backend
- Data is logged in database as well as a csv file

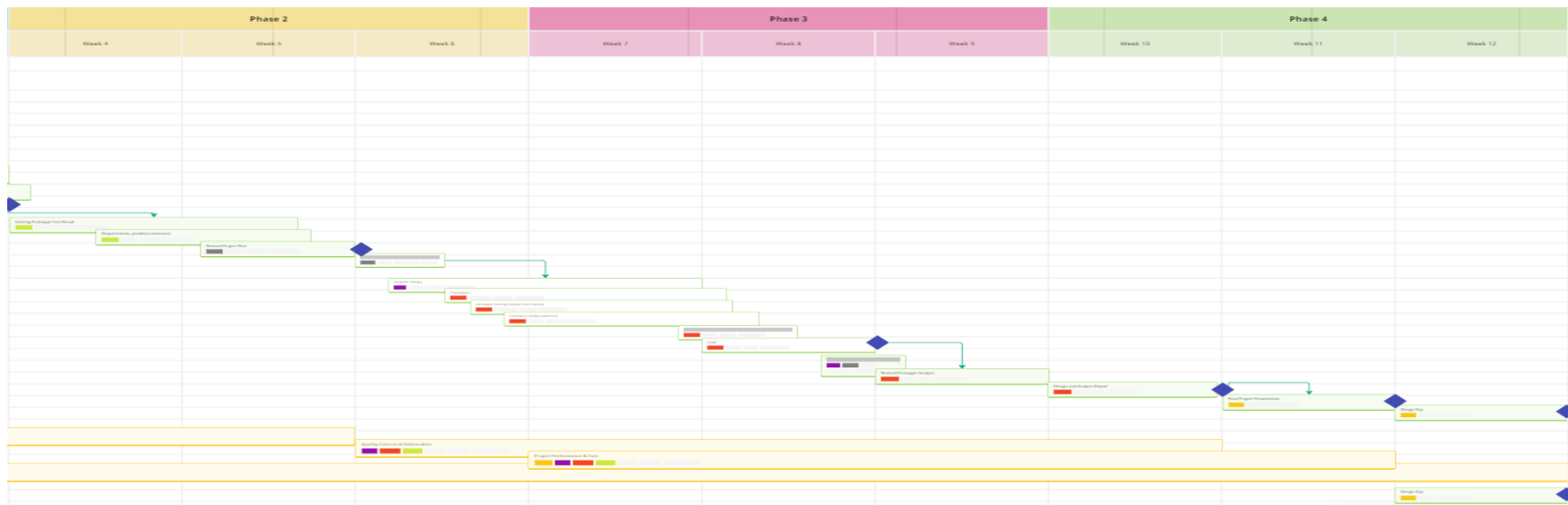


UI and Operations (Web App)



Project Management Skeleton

- Compile all the subtasks together to complete the project milestones
- Client feedback on each phase of the project development
- Ensure the project progress as per the project management skeleton consistently

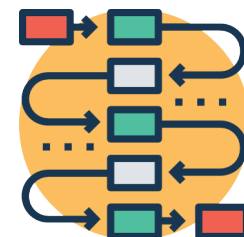


Future Work

- Host the application on a server so it's universally accessible
- Addition of other pollutant sensors
- Reduce the size of the module by using a custom PCB
 - Use a surface mounted microcontroller with custom embedded C code to lower power consumption

Learning Outcome

- Understanding client need specifications and problem definition are the most important tasks before starting any new projects
- Consulting with experts/similar disciplinary people at the planning & design phase can help integrating new ideas or features
- Proper project planning and consistent feedback on the progress can keep the teams work on track



Any questions?

(Please don't ask us any questions we want this to be over)

Thank you!